

Ch-1: Preferences and utility

① What is preference based approach of consumer's behavior? Explain and interpret the assumption of preference based approach. (e.g. completeness, Reflexivity, Rationality)

Ans:

Preference Based Approach: analyzing how the individual uses his preferences to choose an element(s) from the set of alternatives X .

Preferences: "attitudes" of the decision-maker towards a set of possible alternatives X . These attitudes could be obtained by presenting a questionnaire to the individual in which we ask: for any pair of elements x and y in the set of alternatives, X , namely for all $x, y \in X$, how do you compare element x and y ? Check one and only one box.

- ☐ I prefer x to y ($x \succ y$)
- ☐ I prefer y to x ($y \succ x$)
- ☐ I am indifferent ($x \sim y$)

Note that we are asking the individual to check on box. This is related to with the completeness assumption on strict preference.

By asking

check one box
(i.e. not refrain
from answering)

check only one
~~box~~ box.

0

Don't add any new box
in which the individual
says, "I love x and hate y ".

We impose the assumption

Completeness: individuals must compare
any two alternatives, even the ones they
don't know.

The individual is capable of comparing
any pair of alternatives.

We don't allow the individual to
specify the intensity of his preferences

Completeness:

For any pair of alternatives $x, y \in X$, the
individual decision maker:

☐ $x \succ y$ or

☐ $y \succ x$ or

☐ both i.e. $x \sim y$

The decision maker is allowed to choose one and
only one of the above boxes.

Limitation/Remark about completeness: Not all binary relation
satisfy completeness. A binary relation is a ordered pair
pairs (x, y) from a set $x, y \in X$ such as the two
elements that the individual compared in the questionnaire
shown above (e.g. two bundles of goods).

For example: ① Binary relation "is the brother of" does not ^{satisfy} ~~satisfy~~

② If we select Bob and John from this group X , we might observe that neither John is the brother of Bob nor Bob is the brother of John, meaning they are not related. That is, not all pairs of alternative are comparable according to this binary relation, implying that binary relation does not satisfy completeness.

[In short, — "is the brother of": John $\nrightarrow$ Bob and Bob $\nrightarrow$ John if they are not brothers.]

③ Similarly, the binary relation "to be the father of" does not satisfy completeness, since from a group of people, we can select two individual who are not related.

[In short, — "is the father of": John $\nrightarrow$ Bob and Bob $\nrightarrow$ John if the two individuals are not related.]

Not all pairs of alternatives are comparable according to these two relations.

Weak Preferences :

In order to learn weak preferences of an individual consider the following questionnaire :

For all $x, y \in X$, where x and y are not necessarily distinct, is x at least as preferred to y ?

- Yes, $(x \succeq y)$
- No $(y \succeq x)$

Respondents must answer yes, no or both, where checking both boxes reveals that the individual is indifferent between alternative x and y , $x \sim y \in X$.

Note that the above statement relates to completeness, but in the context of weak preference $\succeq$ rather than strict preference $>$.

Reflexivity

Reflexivity: every alternative x is weakly preferred to, at least, one alternative; itself.

A preference relation satisfies reflexivity if for any alternatives $x \in X$, we have that:

- ① $x \sim x$: any bundle is indifferent to itself.
- ② $x \succeq x$: any bundle is preferred or indifferent to itself.
- ③ $x \not\prec x$: any bundle belongs to at least one indifference set, namely, the set containing itself if nothing else.

Reflexivity is implied from completeness. Indeed if we replace alternative y for x , we can transform the assumption of completeness into the assumption of reflexivity.

Rationality:

A preference relation $\succeq$ is rational if it satisfies the following two properties:

Completeness: For any pair of alternatives, $x, y \in X$, either $x \succeq y$ or $y \succeq x$ or both ($x \sim y$).

Transitivity: For any bundle three alternatives, $x, y, z \in X$, if $x \succeq y$ and $y \succeq z$, then it must be that $x \succeq z$.

Example 1: checking for completeness and transitivity:

Consider an individual with preferences

Completeness:

① $x \succeq y$ if and only if $\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i$ or

② $y \succeq x$ if and only if $\sum_{i=1}^M y_i \geq \sum_{i=1}^M x_i$ or

③ both $x \sim y$ if $\sum_{i=1}^N x_i = \sum_{i=1}^N y_i$.

• Transitivity:

① If $x \succeq y$, $\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i$ and

② $y \succeq z$, $\sum_{i=1}^N y_i \geq \sum_{i=1}^N z_i$

③ Then it must be that $\sum_{i=1}^N x_i \geq \sum_{i=1}^N z_i$ (which

implies $x \succeq z$, as required

The assumption of transitivity is understood as that preferences should not cycle.

Example violating transitivity:

apple $\succeq$ banana and banana $\succeq$ orange

apple $\succeq$ orange by transitivity

but orange $>$ apple

[$\Leftrightarrow$ equivalent to]

Q-2 Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity.

Ans:

Utility Function :

A function $u : X \rightarrow \mathbb{R}$ from the set of alternatives to the set of real numbers is a utility function representing a preference relation $\succsim$ if, for every pair of alternatives $x, y \in X$,

$$x \succsim y \text{ is equivalent to } u(x) \geq u(y)$$

Let us emphasize two points in this definition.

① Only the ranking of alternatives matters.

- That is, it does not matter if

$$u(x) = 14 \text{ or if } u(x) = 2000$$

$$u(y) = 10 \text{ or if } u(y) = 3$$

We do not care about cardinality (the number the utility function associates with each alternative) & instead care about ordinality (ranking of utility but values among alternatives).

② If we apply any strictly increasing function $f(\cdot)$ on $u(x)$, i.e.,

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ such that } v(x) = f(u(x))$$

The new function keeps the ranking of alternatives intact and therefore, the new function still represents the same preference relation.

Example :

$$v(x) = 3 u(x)$$

$$v(x) = 5 u(x) + 8$$

Desirability

We consider that x is an N -dimensional bundle, $x \in \mathbb{R}^N$, that is, $x = (x_1, x_2, \dots, x_N)$, where its k -th component represents the amount of good (or service) k , $x_k \in \mathbb{R}$

Monotonicity : A preference relation satisfies monotonicity if, for any two bundles, $x, y \in X$, where $x \neq y$

$x_k \geq y_k$ for every good k , implies $x \succeq y$ and

$x_k > y_k$ for every good k , implies $x \succ y$.

That is, increasing the amounts of some commodities (without reducing the amount of any

other commodity) can not hurt, $x \succeq y$ and increasing the amounts of all commodities is strictly preferred, $x \succ y$.

Strong Monotonicity: A preference relation satisfies strong monotonicity if, for any two bundles $x, y \in X$ where $x \neq y$ and $x_k \geq y_k$ for every good k , we have that $x \succ y$ and $x_l \geq y_l$ for at least one good l .

That is, even if we increase the amount of only one of the commodities, we make the consumer strictly better off.

Q-3: Show that strong monotonicity $\Rightarrow$ monotonicity

Relationship between monotonicity and utility function

Monotonicity in preferences implies that the utility function is weakly monotonic (weakly increasing) in its arguments.

That is, increasing some of its arguments weakly increases the value of the utility function and increases all its arguments strictly increases its value.

More formally, for any scalars $\alpha > 1$,
monotonicity in preferences implies that

$$u(\alpha x_1, x_2) \geq u(x_1, x_2)$$

$$u(\alpha x_1, \alpha x_2) > u(x_1, x_2)$$

where, $\alpha > 1$ implies indicates that we increase the amount of each good by a common factor α .

However, strong monotonicity in preferences implies that the utility function is strictly increasing in all its arguments. That is, if we increase some of its arguments (even only one argument), the value of the utility function strictly increases.

More formally, strong monotonicity in preferences implies that for $\alpha > 1$,

$$u(\alpha x_1, x_2) > u(x_1, x_2)$$

where we have only increased the amount of one of the goods, x_1 .

Let us check if the following utility functions represent an underlying preference relation that satisfies monotonicity alone or strong monotonicity.

Example:-1:

$u(x_1, x_2) = \min \{ax_1, bx_2\}$, where $a, b > 0$,
representing that good 1 and 2 are regarded as
perfect complements.

→ It satisfies monotonicity since, if we increase all
arguments by a common factor $\delta > 0$, we obtain

$$\min \{ax_1 + \delta, bx_2 + \delta\} > \min \{ax_1, bx_2\}$$

→ This result does not satisfy strong monotonicity,
since, if we increase one argument alone (eg. the
amount of good 1 consumed), we obtain

$$\min \{ax_1 + \delta, bx_2\} \nless \min \{ax_1, bx_2\}$$

$\nless = \text{not greater than}$

This happens, when $\min \{ax_1, bx_2\} = bx_2$, since in this
the utility level is unaffected if we increase the amount of good
without simultaneously increasing the amount of good 2.

Or, this example can be written as:

$$u(x_1, x_2) = \min \{x_1, x_2\}$$

→ Monotone, since

$$\min \{x_1 + \delta, x_2 + \delta\} > \min \{x_1, x_2\} \text{ for all } \delta > 0.$$

→ Not strongly monotone, since

$$\min \{x_1 + \delta, x_2\} \nless \min \{x_1, x_2\}$$

$$\text{if } \min \{x_1, x_2\} = x_2$$

Example: -2

$U(x_1, x_2) = x_1 + x_2$, good 1 and good 2 as perfect substitute.

- Monotone, since

$$(x_1 + \delta) + (x_2 + \delta) > x_1 + x_2 \text{ for all } \delta > 0.$$

⇒ It satisfies strongly monotonicity, since

$$(x_1 + \delta) + x_2 > x_1 + x_2$$

Hence, strong monotonicity implies monotonicity.

Strong monotonicity $\Rightarrow$ monotonicity.

but the converse is not necessarily true.

Q-4: Show that strong monotonicity $\Rightarrow$ Monotonicity $\Rightarrow$ LNS

Non-Satiation (NS):

A preference relation satisfies NS if, for every $x \in X$, there is another bundle in set X , $y \in X$, which is strictly preferred to x , i.e. $y \succ x$.

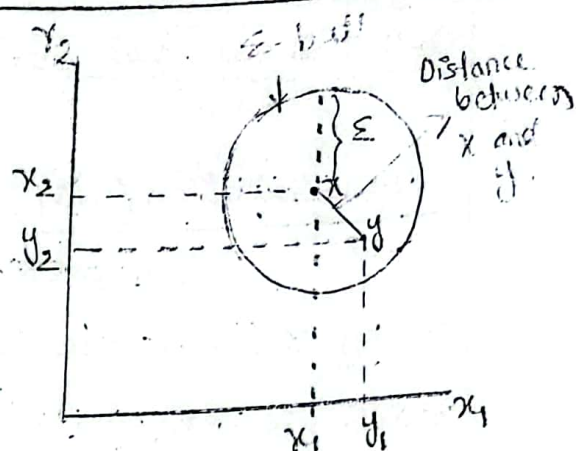
NS is too general, since we could think about a bundle y containing extremely larger amounts of some goods than x .

Local non-Satiation (LNS):

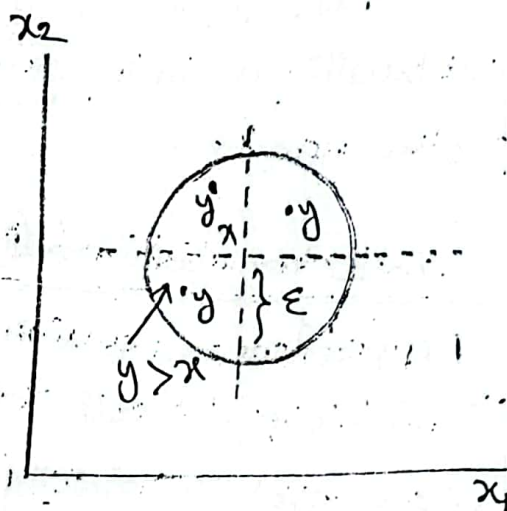
A preference relation satisfies local non-satiation (LNS) if, for every bundle $x \in X$, we can find another bundle $y \in X$ that is strictly preferred to x ($y \succ x$) and where y is less than ε -away from x , $\|y - x\| < \varepsilon$ for every $\varepsilon > 0$.

$\|y - x\| = \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2}$ is the Euclidean distance between x and y , where $x, y \in \mathbb{R}_+^2$. In words, for every bundle x and for every distance ε from x , we can find a more preferred bundle y .

- A preference relation satisfies $y \succ x$ even if bundle y contains less of good 2 (but more of good 1) than bundle x .



- A preference relation satisfies $y \succ x$ even if bundle y contains less of both goods than bundle x .

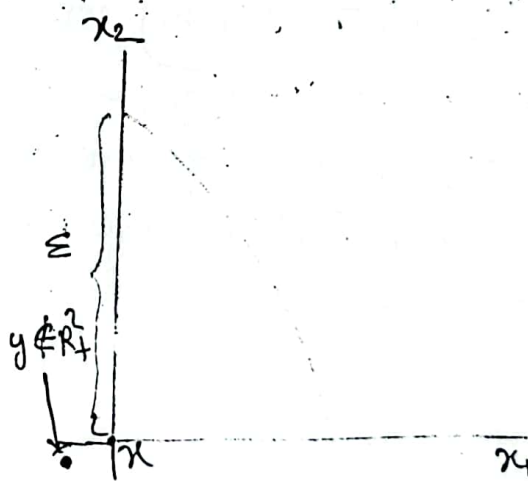


Violation of LNS

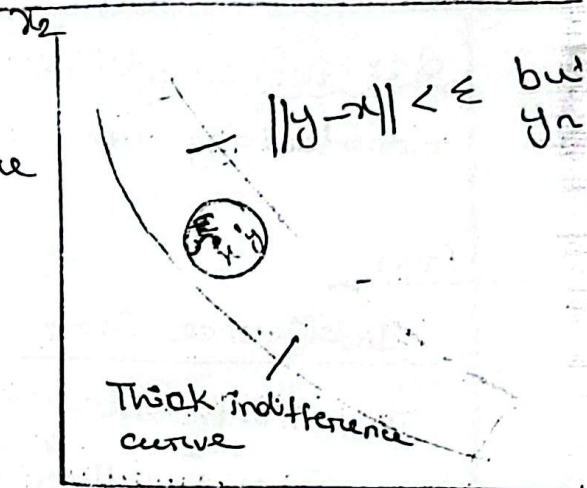
- LNS rules out the case in which the decision maker regards all goods as bads.

- Although $y \succ x$, y is unfeasible given that it lies away from the consumption set, i.e.,

$$y \notin \mathbb{R}_+^2$$



- LNS also rules out thick indifference sets. Bundles y and x lie on the same indifference curve. Hence decision maker is indifferent between x and y , i.e. $y \sim x$.



→ Ans: 4 → Question 4-3 of answer sheet, then follow the same.

Note that, if a preference relation satisfies monotonicity, it must also satisfy LNS.

Given a bundle $x = (x_1, x_2)$, increasing all of its components yields a bundle $(x_1 + \delta, x_2 + \delta)$, which is strictly preferred to bundle (x_1, x_2) by monotonicity. Here there is a bundle, $y = (x_1 + \delta, x_2 + \delta)$ such that $y \succ x$ and $\|y - x\| < \epsilon$.

where δ can be made arbitrarily small, $\delta \rightarrow \epsilon > 0$. Therefore monotonic preference relations also satisfy LNS (In addition, since strong monotonic preference relations are also monotonic, strongly monotonic preference relations also satisfy LNS). In summary,

strong monotonicity $\Rightarrow$ monotonicity $\Rightarrow$ LNS.

Q5: Define indifference set. Explain how does strong monotonicity and indifference curve.

Ans:

Indifference Set:

The indifference set of a bundle $x \in X$ is the collection of all ^{other} bundles ^{in set X ,} $y \in X$, such that $y \sim x$.

$$IND(x) = \{y \in X : y \sim x\}$$

The upper-contour set of bundle x is the set of all bundles $y \in X$, such that $y \succeq x$ (y weakly preferred to x)

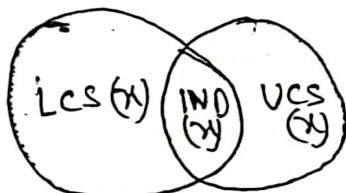
$$UCS(x) = \{y \in X : y \succeq x\}$$

The lower-contour set of bundle x is the set of all bundles $y \in X$, such that $x \succeq y$ (x weakly preferred to other bundle y)

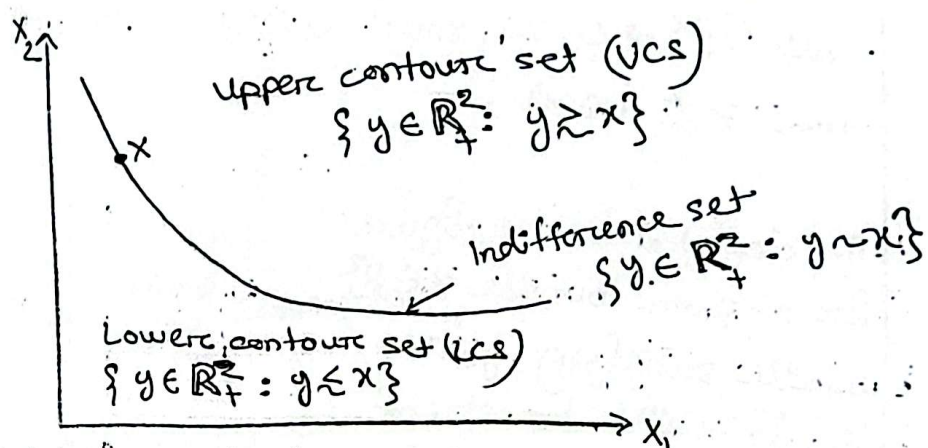
$$LCS(x) = \{y \in X : x \succeq y\}$$

Therefore, $IND(x)$ is the intersection of $UCS(x)$ and $LCS(x)$, that is,

$$IND(x) = UCS(x) \cap LCS(x)$$



The below figure depicts examples of IND, UCS and LCS sets where bundle x lies in $\mathbb{R}_+^2$.



Strong monotonicity and IND(x)

Strong monotonicity implies that indifference curves must be negatively sloped / downward sloped.

In contrast, if an individual preference relation satisfies LNS, indifference curves can be upward sloping.

This can happen, if, for instance, the individual regards good 2 as desirable but good 1 as a bad.

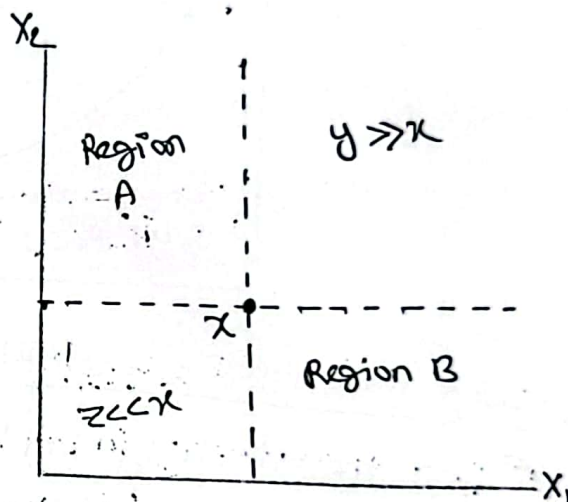
→ To see this, consider a bundle $x \in X$.

By strong monotonicity, other bundles $y \in X$, where $y \gg x$ that contain strictly more of all goods than bundle x does—that is $y \gg x$, must be strictly preferred to x , $y \succ x$, thus implying that $y \notin \text{IND}(x)$.

$[x \in X, y \in X \text{ where } y \gg x, \text{ so } y \succ x, \text{ implies } y \notin \text{IND}(x)]$

Similarly bundles $z \in X$, where $z \neq x$, containing strictly less of every good than bundle x does, that is, $z \ll x$, must satisfy $z \prec x$, thus entailing that $z \notin \text{IND}(x)$.

As depicted in the figure for a given bundle $x \in \mathbb{R}_+^2$ bundle satisfying $y \gg x$ lie to the northeast of x and bundles satisfying $z \ll x$ lie to the southwest of x . Since the indifference set



$\text{IND}(x)$ can not lie on either of these regions by strong monotonicity, the only regions where $\text{IND}(x)$ can lie are regions A and B, which yields a downward sloping indifference curve passing through x .

Alternatively, for indifference curves to be upward sloping the preferences cannot satisfy strong monotonicity, but they can satisfy a milder form of desirability such as local nonsatiation.

Q-6: What are the twin definitions of convexity? In which respect they are different from strong convexity strict examine graphically.

Ans:

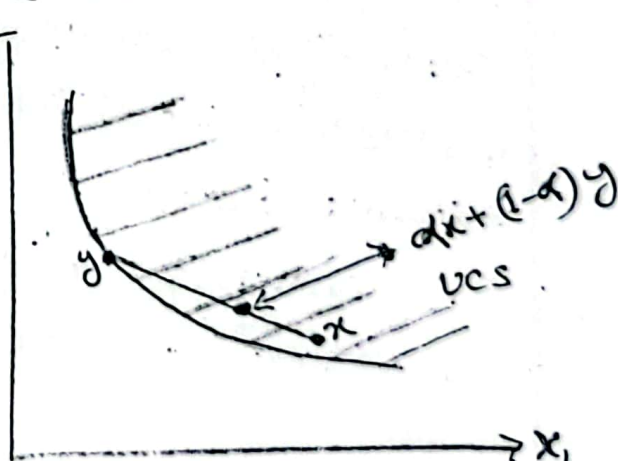
Convexity of Preference

We use two equivalent definitions.

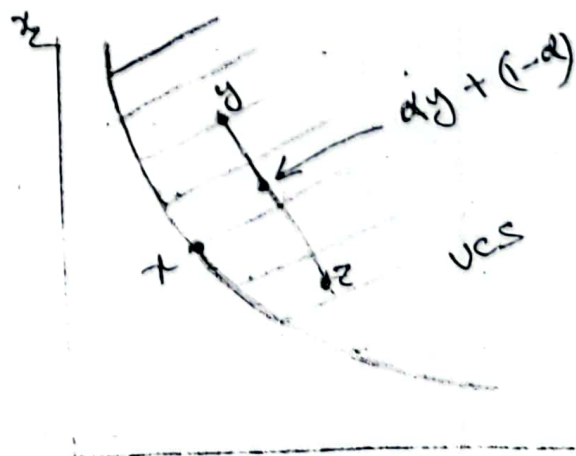
Convexity-1: A preference relation satisfies convexity, if, for any two bundles $x, y \in X$, $x \succeq y$ implies that $\alpha x + (1-\alpha)y \succeq y$ for all $\alpha \in (0,1)$

Intuitively, the convex combination of bundles x and y is weakly preferred to bundle y alone.

The fig. depicts a convex preference relation where, for simplicity, bundle x lies in the upper-contour set of bundle y , $x \in UCS(y)$.



Convexity-2: A preference relation satisfies convexity if, for every bundle x , its upper contour set, $UCS(x) = \{y \in X: y \succeq x\}$, is convex.



That is, for every two bundles y and z ,

$$\begin{cases} y \succeq x \\ z \succeq x \end{cases} \Rightarrow \alpha y + (1-\alpha)z \succeq x$$

for any $\alpha \in (0,1)$

Hence points y, z and their convex combination belong to the UCS of x .

In words, for every two bundles y and z weakly preferred to x , $y \succeq x$ and $z \succeq x$ (which then belong to the UCS of x , as given fig) their convex combination $\alpha y + (1-\alpha)z$ is also weakly preferred to x .

Graphically, convexity in the preference relation implies that the UCS is a convex set and that indifference curves are bowed in toward the origin.

If, in contrast, a preference relation violates convexity, the indifference curves are bowed away from the origin.

Strict convexity

A preference relation satisfies strict convexity, if for every two bundles x and y weakly preferred to z , that is, for all $x, y \in X$, where $x \neq y$ such that $x \succeq z$ and $y \succeq z$, their convex combination $\alpha x + (1-\alpha)y$, where $\alpha \in (0,1)$, is strictly preferred to bundle z .

$$\begin{matrix} x \succeq z \\ y \succeq z \end{matrix} \Rightarrow \alpha x + (1-\alpha)y \succ z, \text{ for all } \alpha \in (0,1)$$

The fig illustrates a preference relation that satisfies strict convexity. When bundles x and y lie on both the UCS and the IND, the definition of strict convexity is satisfied because the convex combination of these two bundles lies strictly inside the UCS, thus implying $\alpha x + (1-\alpha)y > z$.

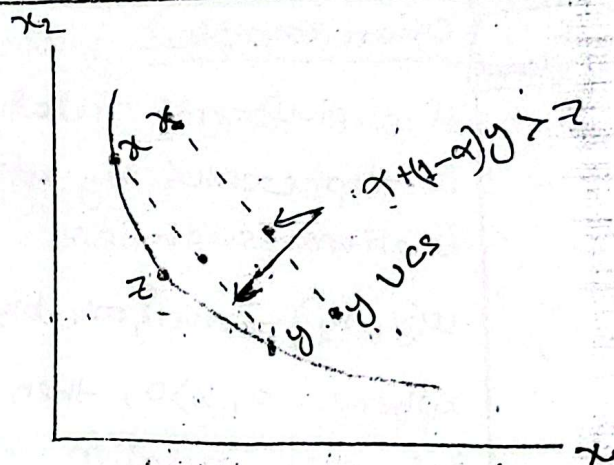


Fig. strictly convex preference.

Convex but not strict convex preferences

Example

For Substitute

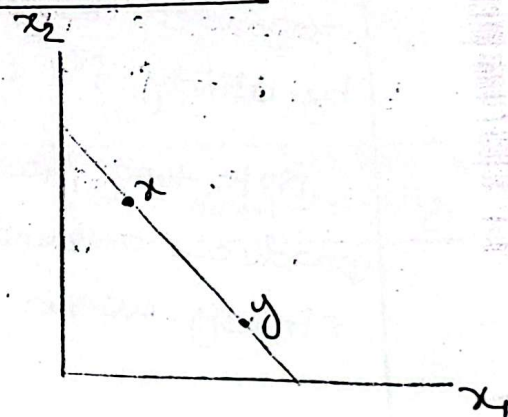
This type of preference relation is represented by linear utility functions such as

$$U(x_1, x_2) = ax_1 + bx_2 \quad \text{where, } a, b > 0$$

where x_1 and x_2 are regarded as substitutes.

or,

In words, Since the convex combination of bundles x and y lies strictly on the indifference curve, their convex combination therefore produces a bundle $\alpha x + (1-\alpha)y$ that the consumer regards as indifferent to bundle z .



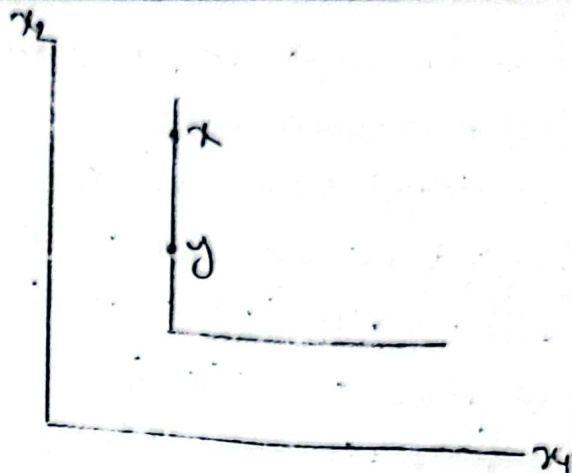
FOC
complement

Other Example :

If a preference relation is represented by utility functions such as

$$u(x_1, x_2) = \min\{ax_1, bx_2\}$$

where $a, b > 0$, then the pref. relation satisfies convexity, but not strict convexity.



or,
Consider preferences over two goods that are regarded as perfect complement, represented by utility u : $u(x_1, x_2) = \min\{ax_1, bx_2\}$

Note that the convex combination of bundles x and y produces a bundle that does not necessarily lie strictly within the VCS.

Q-7: Consider the following utility functions:

① $u(x_1, x_2) = ax_1 + bx_2$

② $u(x_1, x_2) = \min(ax_1, bx_2)$

③ $u(x_1, x_2) = ax_1^{1/2} \times bx_2^{1/2}$

④ $u(x_1, x_2) = ax_1^{1/2} \times bx_2^2$

Examine, which function satisfies convexity, strict convexity or both.

Ans:

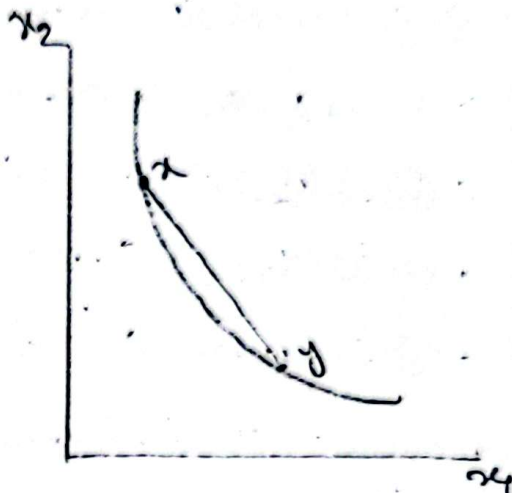
Note: If a preference relation satisfies strict convexity, it must also satisfy convexity; whereas the converse is not necessarily true.

	Satisfies convexity	Satisfies strict convexity
① $ax_1 + bx_2$	✓	X
② $\min(ax_1, bx_2)$	✓	X
③ $ax_1^{1/2} \times bx_2^{1/2}$	✓	✓
④ $ax_1^{1/2} \times bx_2^2$	X	X
Another $ax_1^{1/2} + bx_2^2$	✓	✓
$ax_1^2 + bx_2^2$	X	X

Interpretation of convexity

① Taste for Diversification:

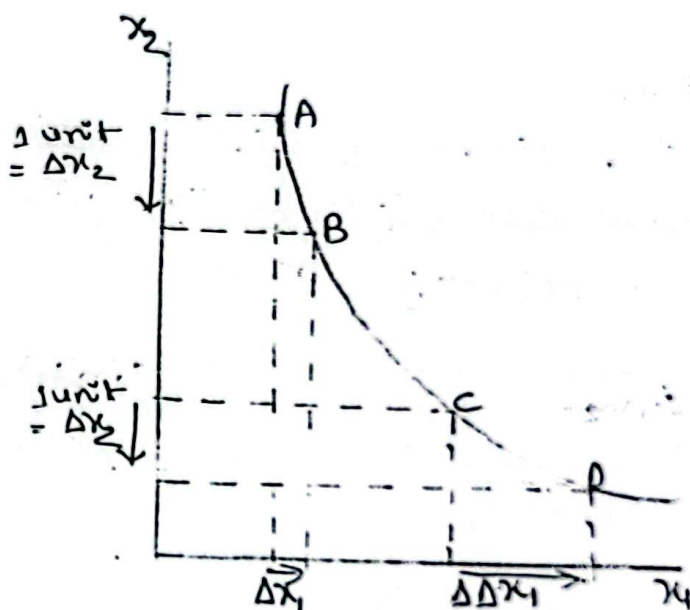
An individual with convex preferences prefers the convex combination of bundle x and y , than either of those bundles alone.



② Diminishing marginal rate of substitution:

$$MRS_{1,2} = \frac{\partial u / \partial x_1}{\partial u / \partial x_2}$$

MRS describes the additional unit amount of good 1 that the consumer needs to receive in order to keep her utility level unaffected, when the amount of good 2 is reduced by one unit.



Hence ~~the~~ a diminishing MRS implies that the consumer needs to receive increasingly larger amounts of good 1 in order to accept further reduction of good 2.

Remark

Let us show that the slope of the indifference curve is given by the MRS.

Consider a continuous and differentiable utility function $u(x_1, x_2, \dots, x_n)$

Totally differentiating, we obtain :

$$du = \frac{\partial u}{\partial x_1} dx_1 + \frac{\partial u}{\partial x_2} dx_2 + \dots + \frac{\partial u}{\partial x_n} dx_n$$

But since we move along the same indifference curve $du = 0$.

Inserting $du = 0$:

$$\Rightarrow 0 = \frac{\partial u}{\partial x_i} dx_i + \frac{\partial u}{\partial x_j} dx_j$$

$$\text{or, } -\frac{\partial u}{\partial x_i} dx_i = \frac{\partial u}{\partial x_j} dx_j$$

If we want to analyze the rate at which the consumer substitutes units of good i for good j ,

we must solve for $\frac{dx_j}{dx_i}$, to obtain

$$-\frac{dx_j}{dx_i} = \frac{\frac{\partial u}{\partial x_i}}{\frac{\partial u}{\partial x_j}} = MRS_{i,j}$$

Q-8: show that

Q-8: convexity of preferences + USC (u) is convex-defn

convexity of preferences

$\Leftrightarrow$ USC (u) is convex $\Leftrightarrow$ u(.) is quasi-concave.

Ans:

A utility function $u(\cdot)$ is quasi-concave if, for every bundle $x \in X$, the set of all bundles for which the consumer experiences a higher utility, i.e., the

$USC(x) = \{y \in X \mid u(y) \geq u(x)\}$ is convex.

Hence, the following three properties are equivalent:

Convexity of preferences $\Leftrightarrow$ USC (x) is convex $\Leftrightarrow$ U(.) is quasiconcave

Alternative definition of quasiconcavity:

A utility function $u(\cdot)$ satisfies quasiconcavity if, for every two bundles $x, y \in X$, the utility of consuming the convex combination of these two bundles,

$u(\alpha x + (1-\alpha)y)$, is weakly higher than the minimal utility from consuming each bundle separately,

$\min \{u(x), u(y)\}$:

$u(\alpha x + (1-\alpha)y) \geq \min \{u(x), u(y)\}.$

The figure illustrates the definition above. In particular, starting from bundles x and y , with

associated utility levels $u(x)$ and $u(y)$, respectively, we can construct a convex combination of the two bundles. If the utility that the consumer derives from consuming this convex combination is larger than from consuming the bundle associated to the lowest utility level (which in this case is bundle y), we say that this individual's utility function satisfies quasi-concavity.

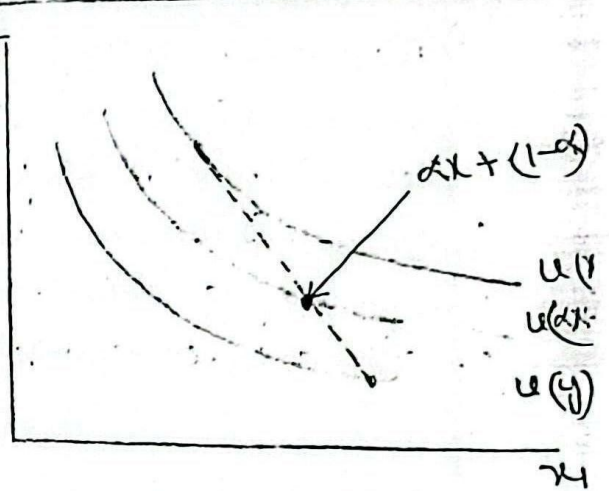


Fig: Quasi-concavity

Strict quasiconcavity

A utility function $u(\cdot)$ satisfies strict quasiconcavity if, for every two bundles $x, y \in X$, the utility of consuming the convex combination of these two bundles,

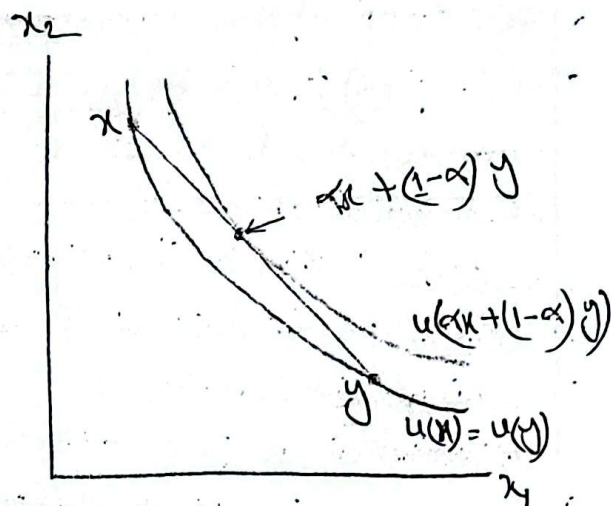
$u(\alpha x + (1-\alpha)y)$, is strictly higher than the minimal utility from consuming each bundle separately,

$$\min \{ u(x), u(y) \} :$$

$$u(\alpha x + (1-\alpha)y) > \min \{ u(x), u(y) \} .$$

utility

If bundles x and y lie on the same indifference curve, then $u(x) = u(y)$. Since indifference curves are strictly convex, $u(\cdot)$ satisfies quasi-concavity.

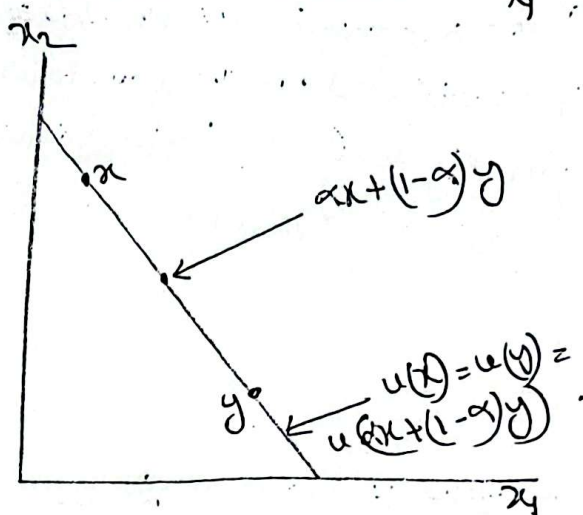


If indifference curves are linear,

$u(\cdot)$ satisfies the definition of a quasi-concavity since

$$u(\alpha x + (1-\alpha)y) = \min \{u(x), u(y)\}.$$

But $u(\cdot)$ does not satisfy strict quasiconcavity.



Q-9: In case of well defined utility function show that
 concavity $\Rightarrow$ Quasi-concavity

Ans:

Relationship between concavity and quasi-concavity of a utility function:

Concavity ^{implies} ~~imply~~ quasi-concavity, but the converse relation is not necessarily true. That is
 concavity $\Rightarrow$ Quasi-concavity.

Hence quasi-concavity is a weaker condition than concavity.

If a function $f(\cdot)$ is concave, it must also be quasi-concave, that is, for any two points $x, y \in X$,

$f(\alpha x + (1-\alpha)y) \geq \alpha f(x) + (1-\alpha)f(y) \geq \min \{f(x), f(y)\}$, for
 for all $\alpha \in (0,1)$.

Where the first inequality follows from the definition of concavity and the second holds true for all ^{concave} ~~concave~~ functions, as depicted in the figure:

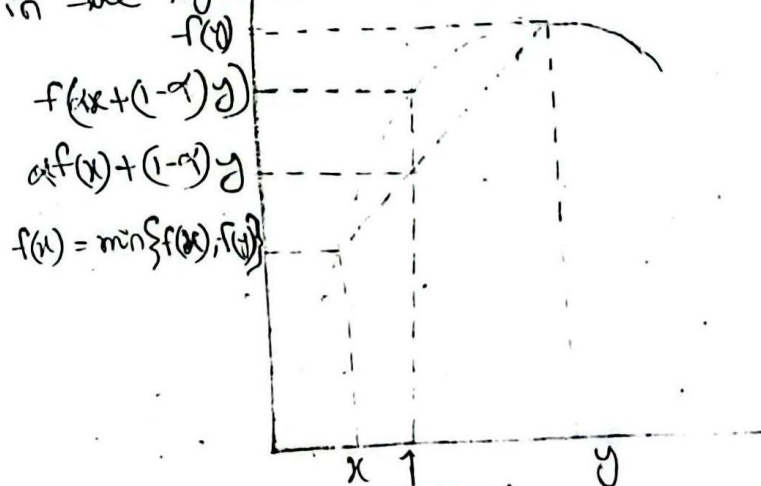


Fig: Concavity implies quasi-concavity.

Elaborating distinct two properties:

② ~~concavity~~

① A concave $U(\cdot)$ exhibits diminishing marginal utility. That is, for an increase in the consumption bundle, the increase in utility is smaller as we move away from the origin.

The "jump" from one indifference curve to another requires:

- a slight increase in the amount of x_1 and x_2 when we are close to the origin.

- a large increase in the amount of x_1 and x_2 as we get further away from the origin.

Concave and quasi-concave utility function, $U(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{4}}$ (Cobb-Douglas)

③ A convex $U(\cdot)$ exhibits increasing marginal utility.

That is, for an increase in the consumption bundle, the increase in utility is larger as we move away from the origin.

The "jump" from one indifference curve to another requires:

- a large increase in the amount of x_1 and x_2 when we are close to the origin, but

- a small increase in the amount of x_1 and x_2 as we get further away from the origin.

Convex but quasiconcave utility function: (~~Cobb-Douglas~~)

$$V(x_1, x_2) = x_1^{\frac{5}{4}} x_2^{\frac{5}{4}}$$

Note:

Utility function $v(x_1, x_2) = x_1^{6/4} x_2^{6/4}$ is a strictly monotonic transformation of $u(x_1, x_2) = x_1^{1/4} x_2^{1/4}$.

That is, $v(x_1, x_2) = f(u(x_1, x_2))$ where $f(u) = u^6$.

Therefore, utility functions $u(x_1, x_2)$ and $v(x_1, x_2)$ represent the same preference relation. Both utility functions are quasiconcave although one of them is concave and other is convex. Hence, we normally require utility functions to satisfy quasi-concavity alone.

Q-10: What are the typical utility functions used in Economic applications. Write down their functional form and calculate and interpret the notion of MRS.

Ans:

We ~~brief~~ briefly list some of the typical utility functions used in economic applications and their functional form:

Cobb-Douglas Utility Functions:

In the case of two goods, x_1 and x_2 ,

$$U(x_1, x_2) = Ax_1^\alpha x_2^\beta \quad \text{where } A, \alpha, \beta > 0.$$

Applying logs on both sides

$$\log u = \log A + \alpha \log x_1 + \beta \log x_2$$

Hence, the exponents in the original $U(\cdot)$ can be interpreted as elasticities:

$$\begin{aligned} \epsilon_{u, x_1} &= \frac{\partial U(x_1, x_2)}{\partial x_1} \cdot \frac{x_1}{U(x_1, x_2)} \\ &= \alpha Ax_1^{\alpha-1} x_2^\beta \cdot \frac{x_1}{Ax_1^\alpha x_2^\beta} \\ &= \alpha. \end{aligned}$$

Intuitively, a one percent increase in the amount of good x_1 increases individual utility by α percent.

Similarly, $\epsilon_{u, x_2} = \beta$.

Special cases:

① If $\alpha + \beta = 1$, $u(x_1, x_2) = Ax_1^\alpha x_2^{1-\alpha}$.

② If $A = 1$, $u(x_1, x_2) = x_1^\alpha x_2^\beta$.

③ If $A = \alpha = \beta = 1$, $u(x_1, x_2) = x_1 x_2$.

Marginal utilities:

$$\frac{\partial u}{\partial x_1} > 0 \quad \text{and} \quad \frac{\partial u}{\partial x_2} > 0$$

Positive marginal utilities satisfying strong monotonicity the Cobb-Douglas utility function yields a diminishing MRS

Diminishing MRS, since

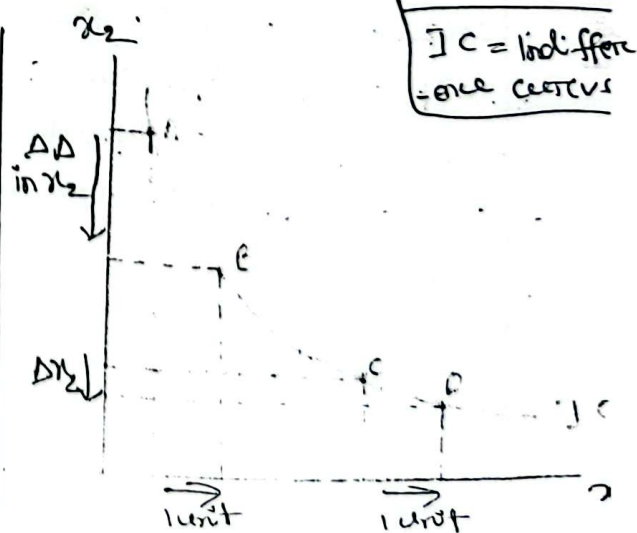
$$MRS_{x_1, x_2} = \frac{\alpha A x_1^{\alpha-1} x_2^\beta}{\beta A x_1^\alpha x_2^{\beta-1}} = \frac{\alpha x_2}{\beta x_1}$$

which is decreasing in x_1 .

Hence, indifference curves become flatter as x_1 increases.

ICs are bowed in toward the origins. Individuals with Cobb-Douglas preferences are willing to give up many units of x_2 in order to increase their consumption of x_1 by one unit when they do not enjoy many units of x_1 (movement from A to B), but such willingness to give up with units of x_2 to obtain one more unit of x_1 decreases

as x_1 becomes relatively abundant (movement from c to d).



Perfect substitutes:

In the case of two goods, x_1 and x_2

utility fn, $u(x_1, x_2) = Ax_1 + Bx_2$, where $A, B > 0$.

Hence, the marginal utility of every good is constant:

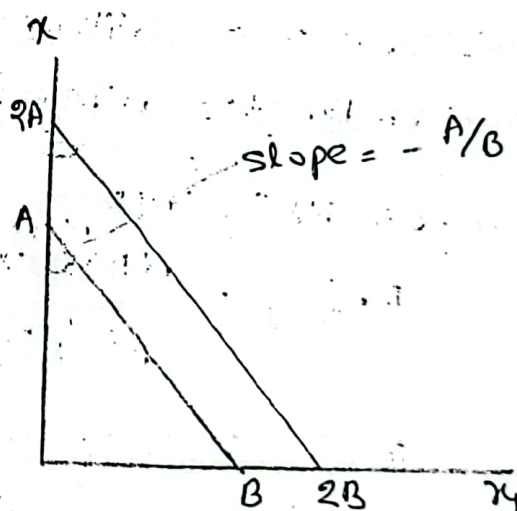
$$\frac{\partial u}{\partial x_1} = A \quad \text{and} \quad \frac{\partial u}{\partial x_2} = B.$$

MRS is also constant, i.e. $MRS_{x_1, x_2} = \frac{A}{B}$.

Therefore, indifference curves are straight lines with a slope of $-\frac{A}{B}$.

Intuitively, the individual is willing to give up A/B units of x_2 in order to obtain one more unit of x_1 and keep his utility level unaffected. Unlike the Cobb-Douglas case, such

willingness is independent in the relative abundance of the two goods. Examples: butter and margarine, coffee and black tea or two brands of unflavored mineral water.



Perfect Complements:

In the case of two goods, x_1 and x_2 ,

$$\text{utility function, } u(x_1, x_2) = A \cdot \min \{ \alpha x_1, \beta x_2 \}$$

where, $A, \alpha, \beta > 0$.

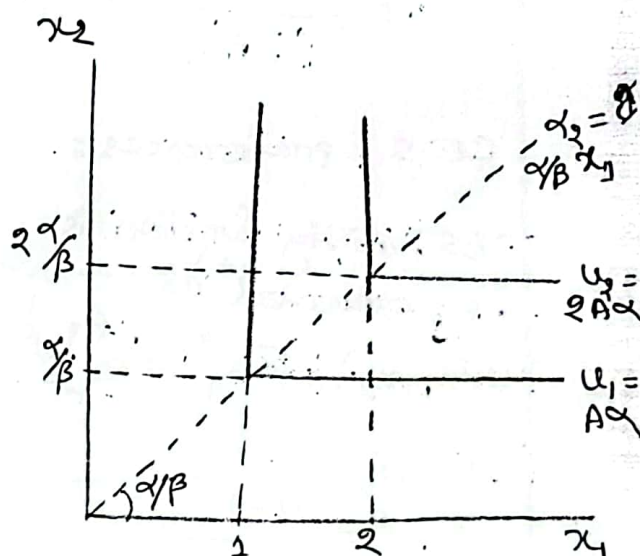
Intuitively, increasing one of the goods without increasing the amount of the other good entails no increase in utility. The amounts of both goods must increase for the utility to go up.

The indifference curve is right angle with a kink at $\alpha x_1 = \beta x_2$.

The slope of a ray

$$x_2 = \frac{\beta}{\alpha} x_1, \quad \frac{\beta}{\alpha} \text{ indicates}$$

the rate at which goods x_1 and x_2 must be consumed in order to achieve utility gains.



Special case: $\alpha = \beta$.

$$\begin{aligned} u(x_1, x_2) &= A \cdot \min \{ \alpha x_1, \alpha x_2 \} \\ &= A\alpha \min \{ x_1, x_2 \} \\ &= B \cdot \min \{ x_1, x_2 \} \text{ if } \beta = A\alpha. \end{aligned}$$

Examples: cars and gasoline or peanut butter and jelly. Other food recipes, which often require the use of ingredients in a fixed proportion, are good examples as well.

Constant Elasticity of Substitution (CES) utility Function:

In the case of two goods, x_1 and x_2 ,

$$u(x_1, x_2) = \left[ax_1^{\frac{\sigma-1}{\sigma}} + bx_2^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

Where, σ measures the elasticity of substitution between goods x_1 and x_2 .

In particular,

$$\sigma = \frac{S\left(\frac{x_2}{x_1}\right)}{S MRS_{1,2}} = \frac{MRS_{1,2}}{\frac{x_2}{x_1}}$$

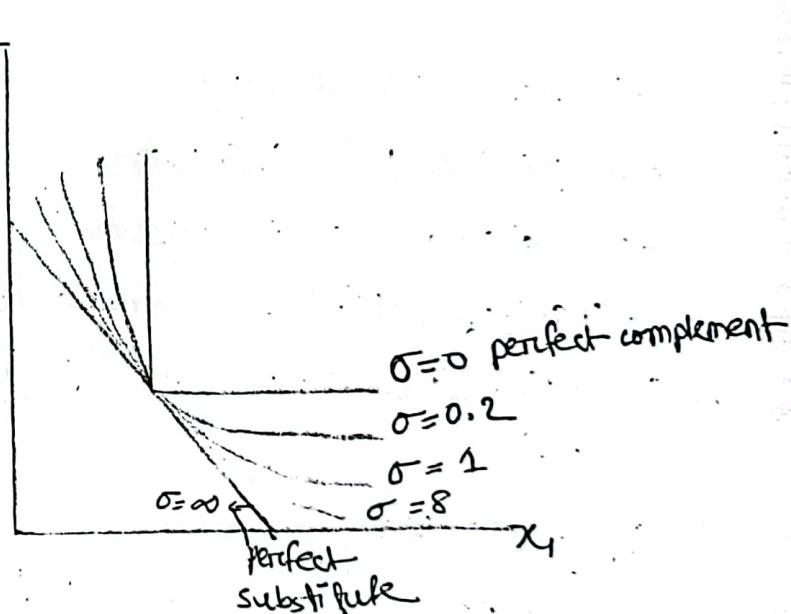
C.E.S. preferences:

CES utility function is often presented as

$$u(x_1, x_2) = [ax_1^\rho + bx_2^\rho]^{\frac{1}{\rho}}$$

where,

$$\rho = \frac{\sigma-1}{\sigma}$$



Quasi-linear Utility Function:

In the case of two goods, x_1 and x_2

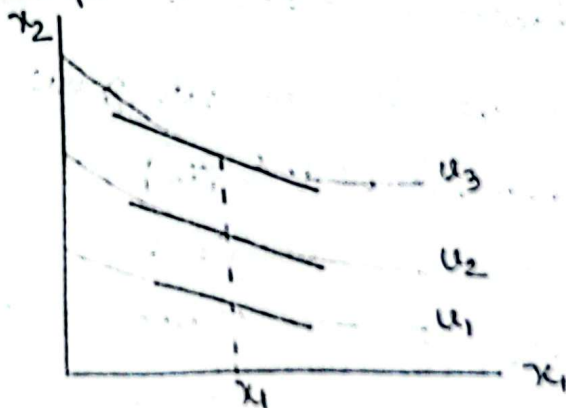
$$U(x_1, x_2) = v(x_1) + b x_2$$

where, x_2 enters linearly, $b > 0$ and $v(x_1)$ is a non-linear function of x_1 .

For example, $v(x) = a \ln x_1$ or $v(x_1) = a x_1^\alpha$ where $a > 0$, $\alpha \neq 1$

The MRS is constant in the good that enters linearly in the utility function (x_2 in our case).

• MRS of quasilinear preferences



For $U(x_1, x_2) = v(x_1) + b x_2$, the marginal utilities are

$$\frac{\partial U}{\partial x_2} = b \quad \text{and} \quad \frac{\partial U}{\partial x_1} = \frac{\partial v}{\partial x_1}$$

which implies, $MRS_{x_1, x_2} = \frac{\frac{\partial v}{\partial x_1}}{b}$

which is constant in the good entering linearly, x_2 .

Quasilinear preferences are often used to represent the consumption of goods that are relatively insensitive to income.

Examples: garlic, toothpaste etc.

Q-11: Examine the following properties of preferences relations:

- i) Homogeneity
- ii) Homotheticity

Homogeneity

A utility function is homogeneous of degree k if varying the amounts of all goods by a common factor $\alpha > 0$ produces an increase in the utility level by α^k .

That is, for the case of two goods,

$$u(\alpha x_1, \alpha x_2) = \alpha^k u(x_1, x_2)$$

where $\alpha > 0$. This allows for:

$\alpha > 1$ in the case of a common increase.

$0 < \alpha < 1$ in the case of a common decrease

Three properties:

① The first-order derivative of a function $u(x_1, x_2)$ which is homogeneous of degree k , is homogeneous of degree $k-1$.

Given,

$u(\alpha x_1, \alpha x_2) = \alpha^k u(x_1, x_2)$, we can show that

$$\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1} \cdot \alpha = \alpha^k \frac{\partial u(x_1, x_2)}{\partial x_1}$$

or rearranging

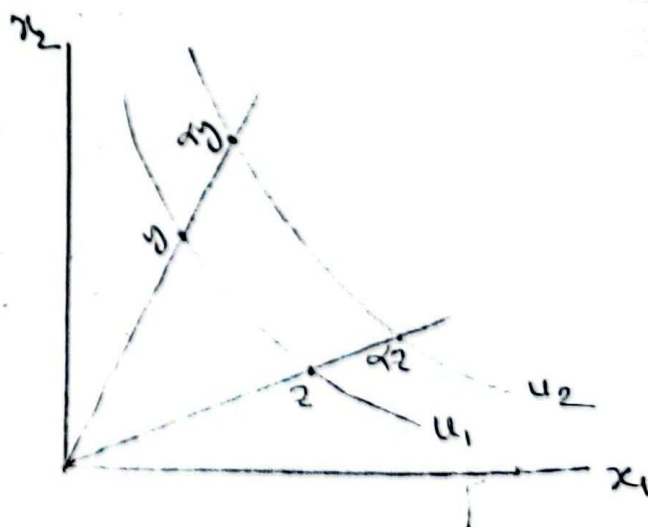
$$u'(\alpha x_1, \alpha x_2) = \alpha^{k-1} u'(x_1, x_2)$$

③ The indifference curves of homogeneous functions are ~~rad~~ radial expansions of the one other.

That is, if two bundles y and z lie on the same indifference curve, i.e., $u(y) = u(z)$, bundles αy and αz also lie on the same indifference curve,

i.e., $u(\alpha y) = u(\alpha z)$. as depicted in the fig

Homogeneous preference:



③ The MRS of a homogeneous function is constant for all points along each ray from the origin.

That is, the slope of the indifference curve at point y coincides with the slope at a "scaled-up version" of point y , αy , where $\alpha > 1$.

The MRS at bundle $x = (x_1, x_2)$ is

$$MRS_{1,2}(x_1, x_2) = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}}$$

The MRS at $(\alpha x_1, \alpha x_2)$ is

$$\begin{aligned} MRS_{1,2}(\alpha x_1, \alpha x_2) &= \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} \\ &= \frac{\alpha^{k-1} \frac{\partial u(x_1, x_2)}{\partial x_1}}{\alpha^{k-1} \frac{\partial u(x_1, x_2)}{\partial x_2}} \\ &= \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}} \end{aligned}$$

where the second equality uses the first property.

Hence, the MRS is unaffected along all the points crossed by a ray from the origin.

Homotheticity

A utility function $u(x)$ is homothetic if it is a monotonic transformation of a homogeneous function.

That is, $u(x) = g(v(x))$, where,

$g: \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function and

$v: \mathbb{R}^n \rightarrow \mathbb{R}$ is homogeneous of degree k .

Properties

① If $u(x)$ is homothetic and two bundles y and z lie on the same indifference curve, i.e., $u(y) = u(z)$, bundles αy and αz also lie on the same indifference curve, i.e., $u(\alpha y) = u(\alpha z)$ for all $\alpha > 0$.

In particular,

$$u(\alpha y) = g(v(\alpha y)) = g(\alpha^k v(y))$$

$$u(\alpha z) = g(v(\alpha z)) = g(\alpha^k v(z))$$

② The MRS of a homothetic function is homogeneous of degree zero.

In particular,

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}}$$

$$= \frac{\frac{\partial g}{\partial v} \cdot \frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial g}{\partial v} \cdot \frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_2}}$$

where, $u(x_1, x_2) = g(v(x_1, x_2))$.

Canceling the $\frac{\partial g}{\partial v}$ terms yields.

$$\frac{\frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\alpha^{k-1} \frac{\partial v(x_1, x_2)}{\partial x_1}}{\alpha^{k-1} \frac{\partial v(x_1, x_2)}{\partial x_2}}$$

Canceling the α^{k-1} terms yields

$$\frac{\frac{\partial v(x_1, x_2)}{\partial x_1}}{\frac{\partial v(x_1, x_2)}{\partial x_2}}$$

In summary,

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}} = MRS_{1,2}(x_1, x_2)$$

Q-12: Proof that if a preference relation satisfies monotonicity and continuity, then there exist a utility function $U(\cdot)$ representing such preference relation.

Ans:

If a preference relation satisfies monotonicity and continuity then there exists a utility function $U(\cdot)$ representing such preference relation.

Proof:

Take a bundle $x \neq 0$.

① By monotonicity, $x \succeq 0$, where $0 = (0, 0, \dots, 0)$.

That is, if bundle $x \neq 0$, it contains positive amounts of at least one good and it is preferred to bundle 0.

② Let $m \equiv \max \{x_1, x_2, \dots, x_n\}$ be the number of units of the most abundant good in bundle x .

Define bundle M as the bundle where all components coincide with the highest component of bundle x . That

$$M = (m, m, \dots, m)$$

Hence, by monotonicity, $M \succeq x$.

③ Bundle 0 and M are both on the main diagonal, since each of them contains the same amount of good x_1 and x_2 .

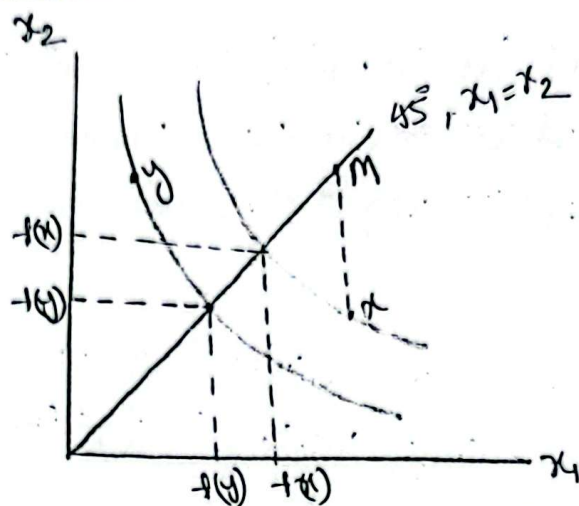


Fig: Existence of a utility function.

By continuity and monotonicity, there exists a bundle that is indifferent to x and which lies on the main diagonal. By monotonicity, this bundle is unique. Otherwise, modifying any of its components would lead to higher/lower indifference curves.

Denote such bundle as

$$(t(x), t(x), \dots, t(x)).$$

Let $u(x) = t(x)$, which is a real number.

Applying the same steps to another bundle $y \neq x$ (see this bundle in the fig), we obtain,

$$(t(y), t(y), \dots, t(y)).$$

and let $u(y) = t(y)$, which is also a real number.

we know that,

$$x \sim (t(x), t(x), \dots, t(x))$$

$$y \sim (t(y), t(y), \dots, t(y))$$

$$x \succeq y$$

Hence, by transitivity, $x \succeq y$ if

$$x \sim (t(x), t(x), \dots, t(x)) \succeq (t(y), t(y), \dots, t(y)) \sim y$$

and by monotonicity,

$$x \succeq y \Leftrightarrow t(x) \geq t(y) \Leftrightarrow u(x) \geq u(y).$$

Note: A utility function can satisfy continuity but still be non-differentiable.

For instance, the Leontief function utility function, $A \min(ax_1, bx_2)$, is continuous but cannot be differentiated at the kink.